Technology Stack

Date	15 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	2 Marks

Define Your Technology Stack:
Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	Creates a responsive and user-friendly interface for entering HDI parameters and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, communicates with the trained machine learning model, and returns predictions efficiently.
3	Database / Data Storage	CSV Dataset (hdi_dataset.csv)	Stores the Human Development Index dataset used for training the machine learning model. A database is not required for this project.
4	Cloud / Hosting / Deployment	Local Flask Server (or Render, if deployed)	Used for local development and testing. Can be deployed to Render for online accessibility.
5	Version Control & CI/CD	Git & GitHub	Git manages version control, while GitHub stores the repository and supports collaboration and project tracking.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Pickle, Jupyter Notebook, VS Code	Scikit-learn trains the Linear Regression model, Pandas and NumPy preprocess and analyze data, Pickle saves the trained model, Jupyter Notebook is used for data analysis, and VS Code is used for application development.